

HINTS AND SOLUTIONS

Q.1 A

Sol. Current at time $t = 0$, $i_1 = \frac{\varepsilon}{R}$

$$\text{Current at time } t = 0, i_2 = \frac{\varepsilon}{R + \frac{18R}{9R}} = \frac{\varepsilon}{3R}$$

$$\therefore i_1 : i_2 = 3 : 1$$

Q.2 B

Sol. $e = \frac{B^2}{2\mu_0}$, so a graph between e and B will be a parabola symmetric about e axis and passing through the origin

Q.3 B

$$\text{Sol. } V_L = \varepsilon_0 e^{-RT/L} = \varepsilon_0 e^{-\ln 4} = \frac{\varepsilon_0}{4}$$

$$\therefore V_R = \varepsilon_0 - V_L = \frac{3\varepsilon_0}{4}$$

Q.4 B

Q.5 B

Sol. Use Kirchoff's current law.

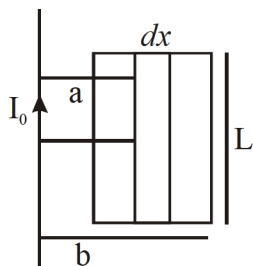
Q.6 B

Q.7 A

$$\text{Sol. } \varepsilon = \int_{-l/4}^{3l/4} B \omega x dx = \frac{1}{4} B \omega l^2$$

Q.8 C

$$\text{Sol. } \int d\phi = \int (dxL) \frac{\mu_0 I}{2\pi x}$$



$$\begin{aligned} \phi &= \frac{\mu_0 I_0 L}{2\pi} \ln \frac{b}{a} \quad \varepsilon_{av} = \frac{\phi}{t} = \frac{4\pi \times 10^{-7} \times 20 \times 20 \times 10^{-2}}{2\pi (.02)} \ln \frac{15}{5} \\ &= \frac{8 \times 10^{-7} \ln 3}{.02} = \frac{8 \times 10^{-7}}{2 \times 10^{-2}} \ln 3 = 4 \times 10^{-5} \ln 3 \end{aligned}$$

Q.9 B

Sol. $B\ell v = \frac{L di}{dt}$ & $\frac{dv}{dt} = -a = -\frac{iLB}{m}$

$$I(t) = \sqrt{\frac{m}{L}} v_0 \sin \omega t; \text{ where } \omega = \frac{B\ell}{\sqrt{mL}} \Rightarrow I = \sqrt{\frac{mv_0^2}{2L}}$$

Q.10 D

Sol. $E = \frac{M di}{dt} = Ma$

$$i = \frac{Ma}{R} (1 - e^{-tR/L})$$

Q.11 C

Sol. Let i_1 is the current in circuit before shifting

$$i_1 = \frac{\varepsilon}{R} \quad \dots (i)$$

Since the flux associated with the inductors will be same just before and just after shifting

$$\therefore i_1 \cdot 5L = i_2 \cdot 9L$$

$$i_2 = \frac{5\varepsilon}{9R}$$

Q.12 C

Sol. $|\int \vec{E} \cdot d\vec{\ell}| = \left| \frac{d\phi}{dt} \right|$

$$E 2\pi \frac{\ell}{2} = \pi \frac{\ell^2}{4} \cdot \frac{dB}{dt}$$
$$E = \frac{\ell}{4} \alpha$$
$$F = \frac{q\ell\alpha}{4}$$

is the electric force on the charge but net force of the particle is zero.

Q.13 BC

Sol. IV should have been : $L = 1H$ and $C = 100\mu F$, $R = 1\Omega$ then answer would have been B & C

(a) V leads I by $\pi/4$ inductive (R-L)

$$R = \omega L \Rightarrow \omega = \frac{R}{L}$$

(b) V lags I by $\pi/4$ capacitive (R-C)

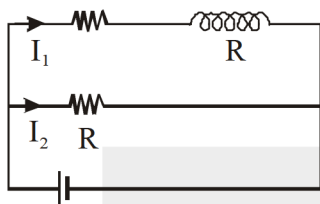
$$R = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{RC}$$

(c) V & I are in same phase $X_L = X_C \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

(d) Not possible in R, L, C

Q.14 C

Sol. $I_1 = \frac{\varepsilon}{R}$; $I_1 = \frac{\varepsilon}{R} \left(1 - e^{-\frac{Rt}{L}} \right) = \frac{\varepsilon}{R} (1 - e^{-1})$



flux linked $= LI_1 = \frac{L\varepsilon}{R} (1 - e^{-1})$

Q.15 AC

Sol. NB At resonance $X_L = X_C$ and $Z = Z_{\min} = R$

$X_L = \omega L$ and $\frac{1}{\omega C} = X_C$

If 'f' is decreased then 'ω' will decrease and hence X_C will increase therefore at $f < f_r$, circuit behaves as capacitive.

V_L and V_C always difference in phase by 180° at any frequency.

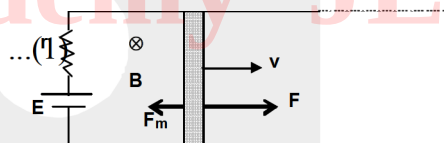
Q.16 [Ans : $800 \mu C$]

Q.17 $\frac{B^2 \ell^2 v_0^2}{2r}$

Sol. (a) Suppose that the speed of the rod is v. The net force acting on the rod is given by

$F - F_m = ma$

where $F_m = i\ell B$ & $i = \frac{(B\ell v - E)}{r}$



$\Rightarrow F_m = \frac{B\ell}{r} [B\ell v - E]$... (2)

Using (1) and (2), we obtain,

$F + \frac{B\ell E}{r} - \frac{B^2 \ell^2 v}{r} = m \frac{dv}{dt}$

The force is given by,

$F = -m v_0 \omega \sin \omega t + \frac{B^2 \ell^2}{r} v_0 \cos \omega t - \frac{B\ell E}{r}$... (3) [3]

The maximum magnitude of the force is

$v_0 \sqrt{(m\omega)^2 + \left(\frac{B^2 \ell^2}{r} \right)^2} + \frac{B\ell E}{r}$... (4) [1]

(b) The power expended by the force is

$F \cdot v = m v_0^2 \omega \sin \omega t \cos \omega t + \frac{B^2 \ell^2}{r} v_0^2 \cos^2 \omega t - \frac{B\ell E}{r} v_0 \cos \omega t$

Its average over a cycle is $\frac{B^2 \ell^2 v_0^2}{2r}$



Q.18 50Ω

Sol. As phase angle between voltage and current is zero hence equivalent impedance of box and capacitor must be resistive $R = 90/3 = 30\Omega$

$$\text{Equivalent } \omega L = \frac{1}{\omega C} = 40\Omega$$

$$\therefore \text{Power factor of the box} = \frac{R}{|Z|} = \frac{30}{\sqrt{30^2 + 40^2}} = \frac{30}{50} = 0.6$$

where $|Z|$ of box $= 50\Omega$

Q.19 $Q_0 = CV \sqrt{[2 - \cos^2 \omega t_0]}$

Sol. For $0 < t < t_0$

Charge on each capacitor is

$$q = CV \cos \omega t \text{ where } \omega = \sqrt{\frac{2}{LC}}$$

and energy of each capacitor is

$$\varepsilon = \frac{CV^2}{2} \cos^2 \omega t_0$$

when one of the capacitor is shorted its energy is lost into heat and remaining energy is

$$= CV^2 - \frac{CV^2}{2} \cos^2 t_0$$

If maximum charge is Q_0

$$\text{then } \frac{Q_0^2}{2C} = CV^2 - \frac{CV^2}{2} \cos^2 \omega t$$

$$\therefore Q_0 = CV \sqrt{[2 - \cos^2 \omega t_0]}$$

Q.20 Emf induced in the circuit at any instant is,

$$\varepsilon = 2 \frac{B_0 x^2 \omega}{2} = B_0 x^2 \omega$$

$$\therefore i = \frac{B_0 x^2 \omega}{R}$$

$$\therefore B \text{ at centroid} = \frac{9\mu_0 i}{2\pi b} \text{ \& } i = \frac{B_0 b^2 \omega}{R}$$

